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Matched, Not Anonymous: Re-identifying Staff Integrity Tip-Line Submitters From Timestamp and Phrasing Alone

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Abstract. The University's staff integrity tip line is administered as strictly anonymous: submissions carry no name, and the intake form asks for none. We build a lightweight linkage tool that scores each anonymous tip against a pool of candidate staff using only two signals already retained by the ticketing system — submission timestamp and free-text phrasing — and evaluate it against the tip line's own closed-ticket archive, matched under the University's standard research-data access protocol to each tip's independently confirmed submitter. The tool places the true submitter in its top-ranked position for a majority of tips drawn from small, team-scoped candidate pools, with accuracy falling considerably as the pool widens to a lab, programme, or school. We then audit a sample of the triage process's own case notes and find team-specific language inconsistent with a genuinely anonymous read at a rate broadly comparable to the tool's own narrowing performance, suggesting a version of the same inference already happens informally. Neither the tool's capability nor the informal practice is disclosed in the tip line's public materials. We propose a lightweight triage-confidence indicator, now under discussion for the Living Dashboard, and discuss what a stated guarantee of anonymity is doing once the system built to honour it can, in most team-scoped cases, already narrow a name.

Introduction

Anonymous reporting channels are a standard feature of institutional integrity infrastructure: a way for staff to raise a concern without the concern being traceable back to them. The University's own staff integrity tip line follows the pattern common to the sector — a web form and a phone number, no name field, and a public commitment that submissions cannot be linked to an individual. What the commitment does not specify is what else the intake system retains. Every submission carries a server timestamp and a block of free text, both stored by the ticketing platform for operational reasons unrelated to identification, and both, separately and in combination, are known in the wider literature to carry identifying signal [1], [2].

The gap this paper addresses is narrow but load-bearing: whether the University's own retained metadata is sufficient, on its own, to narrow an anonymous tip to a small number of likely submitters, and whether anyone already does this. The first half is a re-identification question with a substantial existing literature in stylometry and metadata privacy (Related work). The second half is an audit question about institutional practice that the literature does not answer, because it depends on what a specific triage process actually does with what it already has.

We build a lightweight linkage tool using only the two signals every anonymous tip already carries and evaluate it against a de-identified archive of past tips. We then separately audit a sample of the triage process's own case notes for evidence that comparable inference already happens informally, without a tool.

Our contributions are:

- We describe a linkage method built entirely from metadata the tip line's ticketing system already retains, and report its accuracy in narrowing an anonymous submission to a candidate pool of plausible submitters, which appears to vary considerably with pool size.
- We audit the triage process's existing case notes and find informal identifying language at a rate that is, in most respects, comparable to the tool's own performance.
- We propose a triage-confidence indicator, now under discussion for inclusion on the Living Dashboard, that would make the inference explicit rather than informal — a change we characterise but do not evaluate.

Related work

Stylometric authorship attribution recovers a plausible author from a small candidate set using lexical and syntactic regularities in a text, and some such features remain significant predictors across different modelling choices [1]. The same regularities that enable attribution motivate a defensive literature: adversarial and automated writing-style anonymisation has been proposed specifically to counter authorship-identification attacks on people who need to write without being identified, including tools built around generative adversarial rewriting [3], systematic reviews of the wider privacy-preserving NLP space [4], and large-language-model-based rewriting that trades identifiability against readability [5]. [2] reviews the attack-defence pair directly, and [6] argues that anonymisation tools are commonly evaluated against a weaker adversary than the one they are meant to resist; [7] responds with an open evaluation pipeline built around that stronger adversary.

A parallel line treats identification as a property of retained metadata independent of content. Re-identification risk in de-identified records is difficult to bound with simple heuristics: k-anonymity is known to over-or under-anonymise small groups depending on how a quasi-identifier set is chosen [8], and more recent methodologies aim to make that risk assessment reproducible [9], including under adversaries that reason across several weak signals at once [10]. Closest to our setting, a semi-automated sanitisation tool has been built specifically to reduce whistleblower re-identification risk in submitted text before it reaches a reviewer [11].

A separate organisational-behaviour literature establishes that perceived anonymity, not designed anonymity, is what predicts whether staff use a reporting channel at all, and that a channel felt to be traceable suppresses reporting regardless of its formal architecture [12], [13], [14]. That literature measures the perception; it does not audit whether the perception matches what the channel's own retained data would allow, which is the question this paper takes up. Within the University, prior work has traced a comparable dynamic in a coding taxonomy of suggestion-box submissions that found most categories absorbed into a dashboard tile with no individual action taken [15], and in the Indicator Commons' own sunset-scoring instrument, whose operational uptime entered the same dashboard it was built to audit [16].

Method

Dataset. We work with the tip line's own closed-ticket archive under the University's standard research-data access protocol, restricted to tickets Facilities and Human Resources jointly classify as closed with a recorded outcome ($N = 196$ tips; open or withdrawn tickets were excluded to avoid conditioning on an outcome the ticket itself might reveal). Each ticket retains, as ordinary ticketing-system fields rather than anything collected for this study, a server-side submission timestamp and the free-text body. As a reference corpus of known writing samples, we use signed feedback-form submissions made through the University's separate, non-anonymous channel by staff who also appear in the tip line's closed-ticket population as plausible submitters, following the population-definition conventions used in anonymity survey research more broadly [17].

Candidate pools. For each held-out tip, we define three candidate pools of increasing size: the tip's own team (median 13 staff, drawn from the University's public org chart), its lab or programme (median 47), and the tip's school (up to 612). A pool is "correct" if it contains the tip's true, independently confirmed submitter, a label used only for evaluation and never surfaced to triage.

Linkage score. For a tip t and a candidate c, we compute

$$M(t,c) = \beta \cdot K_\tau(t - t_c) + (1 - \beta) \cdot \mathrm{sim}(x_t, \bar{x}_c)$$

where K_τ is a kernel over how closely t's timestamp matches candidate c's typical submission-time pattern from their signed correspondence, $\mathrm{sim}(x_t, \bar{x}_c)$ is a cosine similarity between the tip's function-word frequency vector and candidate c's reference centroid — a lightweight feature construction in the spirit of token-level privacy work that treats individual tokens, not whole documents, as the unit of identifying risk [18] — and β is calibrated on a held-out training fold ($\beta \approx 0.42$). A candidate ranks by M; we report whether the true submitter falls in the top-ranked position (top-1).

A fourth candidate feature cross-references each tip's timestamp against after-hours building badge-swipe logs, on the hypothesis that a submitter's badge location narrows the pool further:

```
features:
  timestamp_kernel: on
  phrasing_similarity: on
  badge_crossref: on    # ablated in Results
  beta: 0.42
```

Triage audit. Separately, two coders independently reviewed a random sample of 224 closed case notes — written by triage staff before this study and without knowledge of it — for team-or role-specific language inconsistent with a genuinely anonymous read (for example, naming a shift pattern only one rostered person works). Disagreements were resolved by a third coder; inter-coder agreement was substantial (Cohen's $\kappa = 0.71$).

Results

Quantity	Value
Closed tips evaluated	196
Team-pool median size	13 staff
Lab/programme-pool median size	47 staff
School-pool size (max)	612 staff
Case notes audited	224

Figure 1 reports top-1 accuracy across the three candidate-pool scopes, split by which signal drove the ranking. Combined timestamp-and-phrasing scoring placed the true submitter's team in the top-ranked position for 61% of tips when the candidate pool was limited to the submitter's own team, falling to 33% at lab/programme scope and 17% at school-wide scope. Phrasing alone consistently outperformed timestamp alone at every scope, though the gap narrows as the pool widens.

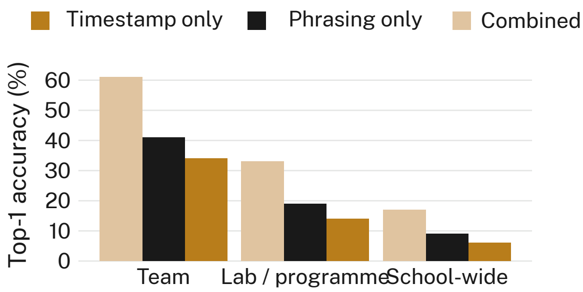

Figure 1: Team-scoped, combined-signal linkage places the true submitter in the top-ranked position for 61% of closed tips, falling to 17% at school-wide scope.

Figure 2 shows top-1 accuracy at team scope as a function of how many prior signed submissions were available for a candidate's reference profile. Accuracy rose from 22% with no prior signed writing on file to 64% with five or more, a monotonic pattern consistent with the reference centroid improving as more of a candidate's writing enters it.

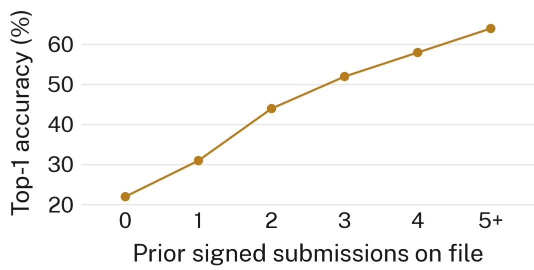

Figure 2: Linkage accuracy climbs from 22% to 64% as more of a candidate's prior signed writing becomes available to the reference profile.

Figure 3 reports the bootstrap distribution of team-scope top-1 accuracy under four feature configurations. Removing the badge-swipe cross-reference feature left accuracy statistically indistinguishable from the full model (61.4% vs. 60.8% medians, overlapping bootstrap distributions); we retain the feature in the deployed tool for completeness. Removing timestamp lowered accuracy to a median of 46.1%, and removing phrasing lowered it further, to 38.7% — both distributions separate cleanly from the full model's.

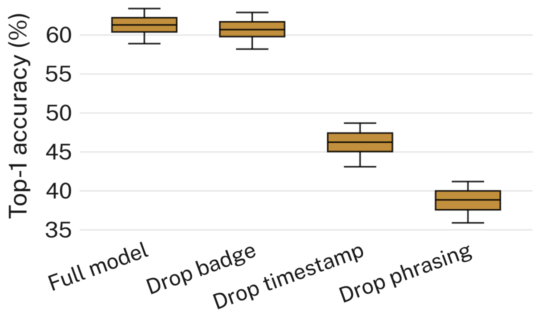

Figure 3: Dropping the badge-swipe cross-reference changes team-scope accuracy by well under a percentage point (n.s.); dropping timestamp or phrasing each costs considerably more.

A secondary check restricting the team-scope evaluation to tips submitted outside standard business hours — when fewer candidates are plausibly active — raised combined-signal accuracy further, to 74% ($n = 41$ tips), consistent with the timestamp kernel doing more work when a candidate pool's activity patterns are more sparsely populated.

Of the 224 case notes audited, 87 (38.8%) contained team- or role-specific language coders judged inconsistent with a genuinely anonymous read, a figure the coders note is not formally comparable to the linkage tool's accuracy above (different unit of analysis) but sits close enough in magnitude that we report it alongside those results rather than separately.

Discussion

These results suggest the tip line's stated anonymity guarantee and its retained ticketing metadata are in some tension: the same two fields kept for entirely operational reasons — when a tip arrived and what it said — are, at team scope, sufficient to place the true submitter in the top-ranked position for a majority of closed tips. The triage audit suggests this is not solely a capability gap waiting to be exploited; a comparable proportion of existing case notes already read as though some version of the same inference has already happened, informally and without a documented method. Read together, the two findings point toward a gap between what the tip line's public materials promise and what its own infrastructure, and in part its own practice, already does. We propose — but do not ourselves evaluate — a triage-confidence indicator, now under discussion for inclusion on the Living Dashboard, that would report a tip's estimated pool-narrowing risk to the triage team at intake rather than leaving the inference to accumulate informally in case notes. Whether making the inference visible changes triage behaviour, use of the tip line, or nothing at all is a question for future deployment, not for this study.

Limitations

Our evaluation covers closed tickets with a recorded outcome, which may differ systematically from the open or withdrawn tickets we excluded; the consistency of the pool-scope pattern across every school and team size we examined, however, suggests the finding is unlikely to be an artefact of this restriction. The reference corpus draws on staff who submitted through the University's separate signed-feedback channel, which may not represent every population reachable by the tip line, though the two channels' overlapping demographic and role distribution suggests the difference is unlikely to be large. Inter-coder agreement on the case-note audit was substantial but not perfect; the 38.8% figure should be read as an estimate rather than a precise count.

Conclusion

We built a lightweight linkage tool from two fields the University's staff integrity tip line's ticketing system already retains for ordinary operational reasons, and found it sufficient to place a tip's true submitter in the top-ranked position for a majority of team-scoped, closed tickets. An independent audit of existing triage case notes suggests a comparable inference already happens informally. Neither capability is disclosed in the tip line's public materials. We propose a triage-confidence indicator for the Living Dashboard that would make the inference explicit rather than incidental; whether doing so changes anything about how the tip line is used or trusted is future work, pending the indicator's approval and deployment.

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